

Head Movement During Five Activities of Daily Living

A single participant smartphone IMU feasibility study

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Abstract

This study tests a methodology and data processing pipeline for recording head range of motion during five activities of daily living (ADL), using a smartphone IMU (Sensor Logger application) mounted on a headband. One healthy adult completed three trials of each: seated desk work, standing desk work, standing phone work, eating, and level walking. Each task was sampled at a requested 50 Hz, referenced by a 10-second neutral hold, and analysed in a 180-second window. Mean flexion-extension ROM ranged from 9.9 to 19.6°; the corresponding coefficient of variation ranged from 22% to 56%. Mean neutral-hold variability was 1.12°, while the initial phone inclination varied by 20.5° across the fifteen task trials, showing variation in sensor placement or reference posture. The study demonstrates a consistent data acquisition and processing workflow. Walking axial rotation is shown with the lap turns. It does not validate head ROM accuracy because no reference measurement was used. One head-mounted IMU estimates head orientation in space; another trunk-mounted IMU is needed to calculate the relative angle between the head and trunk.

Introduction

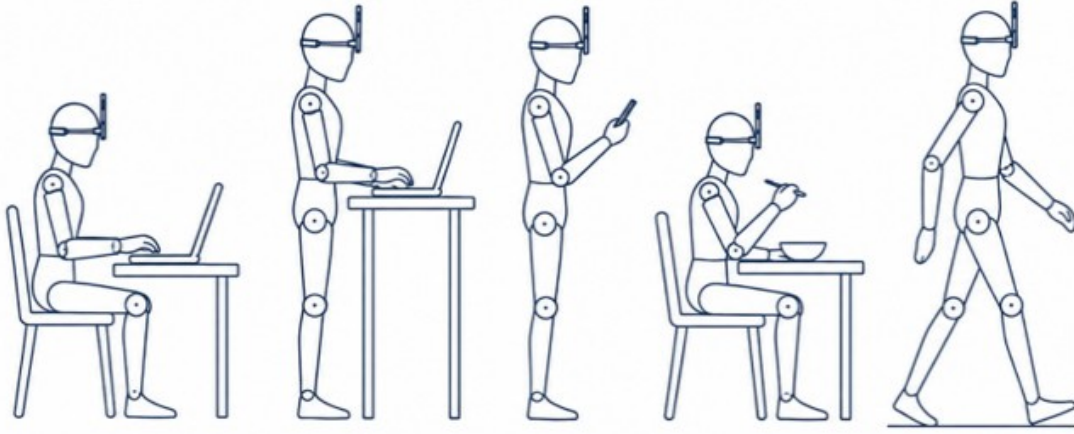
Activities of daily living use different portions of the available head motion range [1]. This is relevant to assistive orthosis devices because functional head movement is a subset of the available range and matters for tasks like screen use, eating, conversation and walking [2]. Inertial measurement units can estimate head orientation in three planes under controlled conditions [3,4], but an IMU attached only to the head estimates head orientation in space without a reference for trunk motion.

The objective of this study is to test the feasibility of recording head movement data for five ADL tasks with three trials each and processing them through a single MATLAB workflow. This is an independent study and does not claim measurement accuracy; it is a proof of methodology.

Methodology

The smartphone IMU was centred on the forehead and held by a strap. A calibration trial was analysed over a 60-second window right after the neutral hold. The calibration movements included extension, flexion, axial rotation in both directions, and lateral bending to both sides, with approximately 10-second holds. The ADL task positions are illustrated in Figure 1(a); Figure 1(b) shows the head reference axes.

(a)



(b)

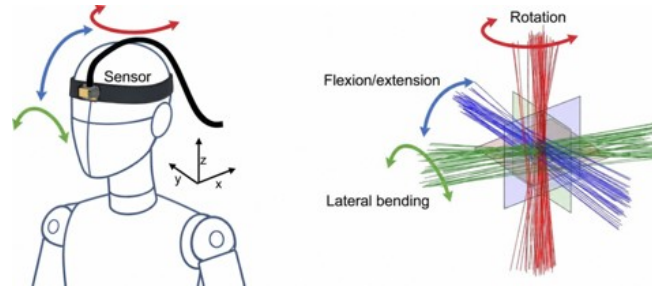


Figure 1. (a) Task positions. (b) Head axes and movement directions. Task mannequins generated with OpenAI.

The smartphone IMU has a built-in gyroscope and accelerometer that provide angular velocities and acceleration in the X, Y and Z directions. The gyroscope and accelerometer data were sampled at a requested 50 Hz and placed on a uniform time grid. The quietest complete 10 s block in the first 30 s was defined as the reference for each trial, also called the neutral hold. Neutral gravity and the phone's z axis defined the head reference frame.

Gravity direction was estimated using a second-order Butterworth low-pass filter at a 0.5 Hz design cutoff, applied forward and backward for zero phase delay. A 0.5 Hz cutoff has been evaluated in accelerometry processing [5]. Head flexion-extension and lateral bending were estimated from gravity in each trial's reference frame. Axial rotation rate was corrected for gyroscope bias using the neutral hold, low-pass filtered at 5 Hz with the same filter order and forward-backward application, and then integrated. Comparable head-motion work used 6 Hz [6]; 5 Hz was selected for this study. The integrated axial angle was linearly detrended to reduce residual drift. A fixed 180-second window began immediately after the neutral hold.

For walking, samples within approximately 1 s of an absolute axial rotation rate higher than 25°/s were excluded from the straight-walking calculations to reduce abrupt changes at lap turns. The complete walking axial rotation trace is shown, including turns. Walking axial ROM is not summarised in the table.

ROM was calculated as the 97.5th percentile minus the 2.5th percentile angle. Trial values were summarised using mean, sample standard deviation, sample variance and coefficient of variation ($CV = 100 \times SD/mean$). For three trial ROM values R_1 , R_2 and R_3 , $mean = (R_1 + R_2 + R_3)/3$; $sample\ SD = \sqrt{[\sum(R_i - mean)^2]/2}$; and $variance = SD^2$. Mean and SD are reported for each available plane; variance and CV describe flexion-extension. Calibration ROM was maximum minus minimum.

Results

The fifteen task recordings had a mean measured sampling rate of 49.92 Hz and mean neutral-hold flexion-extension SD of 1.12°. Initial phone inclination spanned 20.5° across those fifteen recordings. The separate calibration gave 119.6° sagittal ROM. Walking retained 77% of samples after turn exclusion.

Task	Flexion-extension mean $\pm$ SD ($^{\circ}$)	Lateral bending mean $\pm$ SD ($^{\circ}$)	Axial rotation mean $\pm$ SD ($^{\circ}$)	FE variance ($^{\circ 2}$)	FE CV (%)
Seated desk	12.4 ± 7.0	5.2 ± 2.2	6.9 ± 3.7	48.5	56
Standing desk	9.9 ± 3.6	3.6 ± 1.2	6.0 ± 1.4	13.0	36
Standing phone	19.6 ± 8.9	3.8 ± 1.0	6.9 ± 3.3	79.1	45
Eating	13.3 ± 3.0	5.4 ± 2.6	15.1 ± 12.6	8.8	22
Walking	11.7 ± 4.4	3.6 ± 0.5	—	19.1	37

Table 1. Head ROM across three trials per task. FE = flexion-extension. Walking axial ROM is not summarised because lap turns include whole-body rotation; its trace appears below. Variance and CV were calculated before rounding.



Figure 2. Head-angle curves for all three trials of each task, plotted against elapsed time in the 180 s window. Flexion-extension and lateral bending use the neutral reference; axial angles are linearly detrended. Gaps mark excluded walking turns. The walking axial panel retains the complete trace, including turns, and uses a different vertical scale.

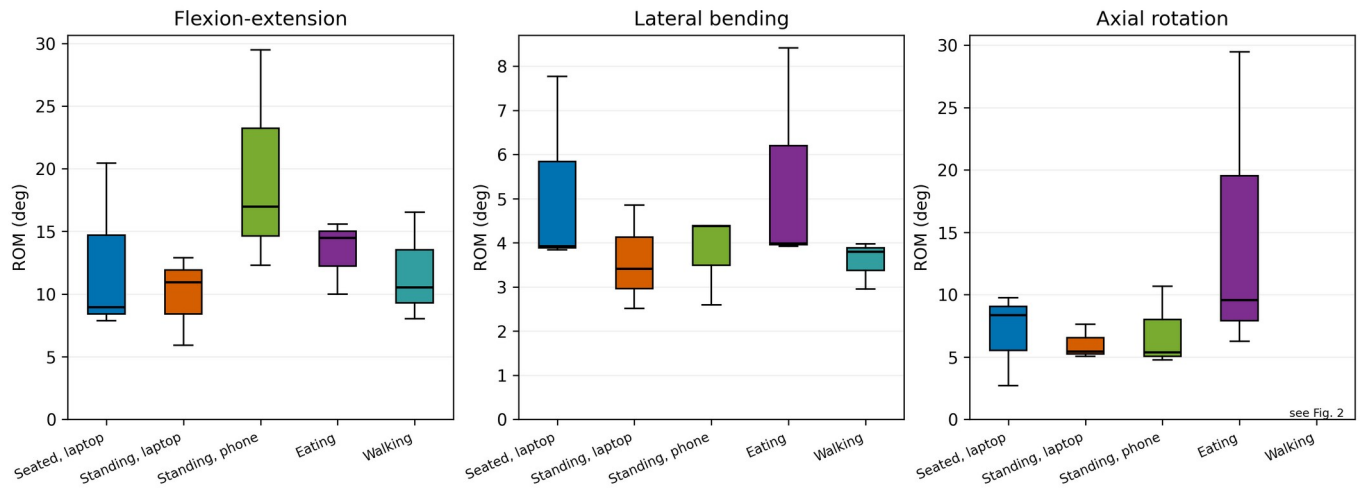


Figure 3. Head ROM across three trials per task: median, quartiles and minimum–maximum.

Discussion

Standing phone use had the largest mean flexion-extension ROM (19.6°). Seated desk work had the highest CV (56%), while eating had the lowest (22%). These values show how much the trial ROM changed between repeats; they do not establish accuracy. The tasks were not scripted closely enough to separate normal changes in behaviour from mounting and processing effects.

Neutral-hold SD includes possible head movement, strap movement and sensor noise. The 20.5° inclination spread suggests changes in mounting or reference posture, but their effect on ROM was not measured separately. Walking remains a useful ADL. Its lap turns explain why the complete axial trace and straight-walking ROM need different interpretations.

Further work should use a repeatable forehead mount and a fixed task script, add a trunk IMU, and compare the estimates with a reference measurement. More participants and repetitions would be needed to assess repeatability beyond this participant.

Limitations

This study used one participant, three trials per task and no reference instrument. A single head sensor cannot isolate head movement relative to the trunk. The 0.5 Hz gravity filter attenuates faster movement, and the filter settings were not validated for this phone. Linear detrending can remove genuine sustained rotation as well as drift. The selected neutral block may differ from the intended hold. These limits prevent clinical or population-level claims.

Conclusion

The workflow processed all fifteen task recordings and produced head ROM estimates across five ADLs. Mean flexion-extension ROM was 9.9–19.6°, with CVs of 22–56%. The method is feasible for recording and inspecting movement, but the present results are not validated reference values.

References

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